

TASK 8 SUBMISSION

Competitor Demand Prediction Using Job Postings

Company Similarity Scoring

How alike are the four companies?

Prepared by	Nayab Khalid
Company specialism	NVIDIA
Task	Task 8 of 10 - Company Similarity Scoring
Method	Canonical skill share vectors, TF-IDF weighted, cosine similarity
Date	20 September 2026
Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Headline: similarity cannot be measured from job titles

Title-based skill vectors carry about one skill per posting, which is too sparse for cosine similarity to mean anything. Every pair scores between 0.02 and 0.22, and the ranking flips depending on which taxonomy is used: NVIDIA's closest peer is Microsoft on one and Meta on the other. The full-text matrix, available for three of the four companies, scores 0.38 to 0.57 and is stable. NVIDIA cannot appear in it, because its source carries no description text.

1 Method, and why there are three matrices

Per the shared analysis config: canonical skill share vectors, TF-IDF weighted, compared with cosine similarity. TF-IDF matters here - a skill every company lists carries no discriminating information, so it is down-weighted.

Task 6 established a bias that this task cannot ignore. The 34 skills added to the taxonomy in Task 4 were chosen by reading NVIDIA job titles, and they lift NVIDIA's coverage from 21% to 72% while lifting the others by 2 to 6 points. A single matrix built on that taxonomy would report that NVIDIA is unlike everyone else, when part of what it reports is that the taxonomy was written by NVIDIA's analyst. So the same calculation is run three ways, and the disagreement between them is the result.

Matrix	Taxonomy	Text	Companies
M1	extended (93 skills)	job titles	all four
M2	pre-extension (59 skills)	job titles	all four - the unbiased control
M3	extended (93 skills)	full descriptions	Google, Meta, Microsoft - NVIDIA has none

2 The two title matrices disagree

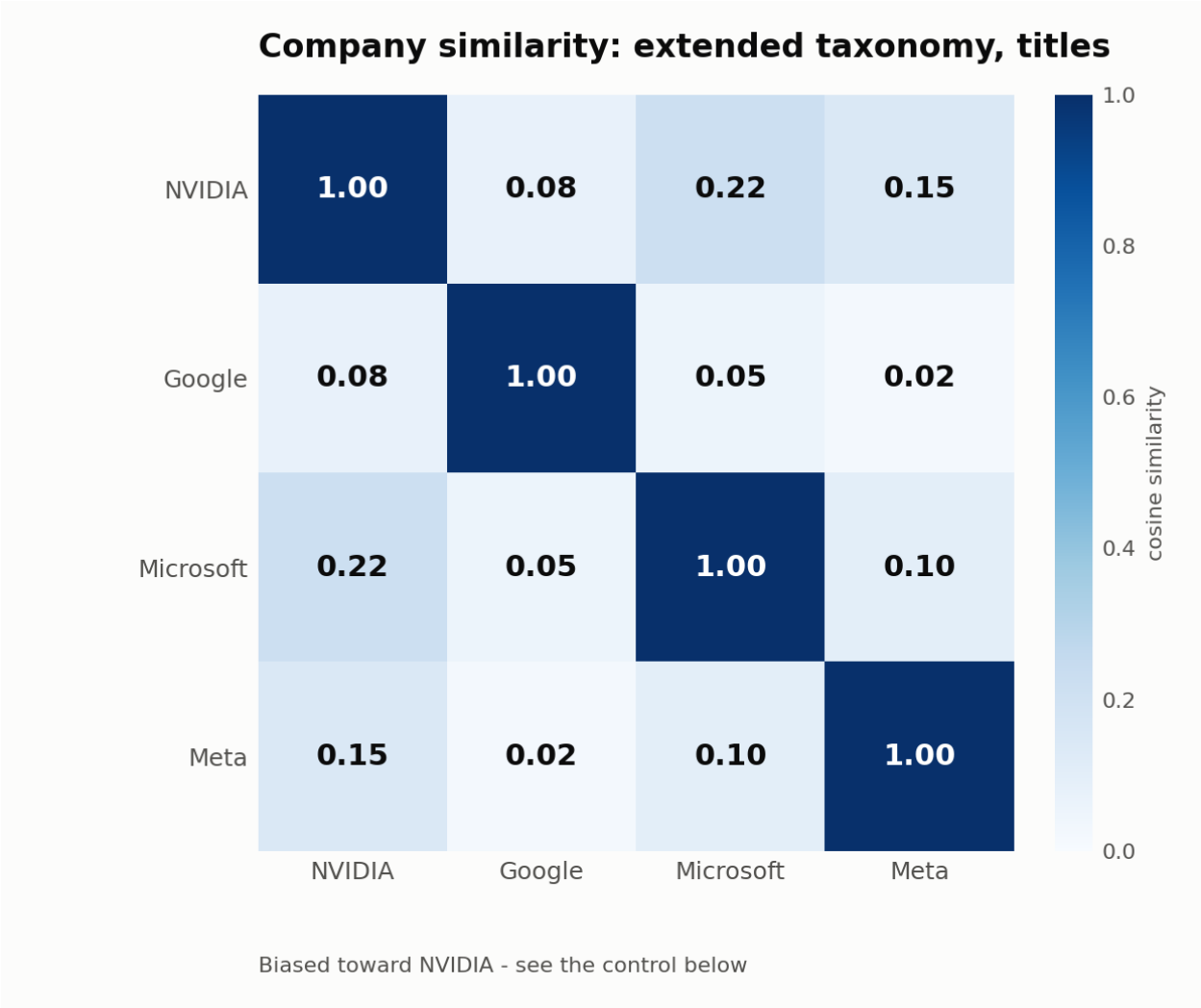


Figure 1. M1 - extended taxonomy, titles.

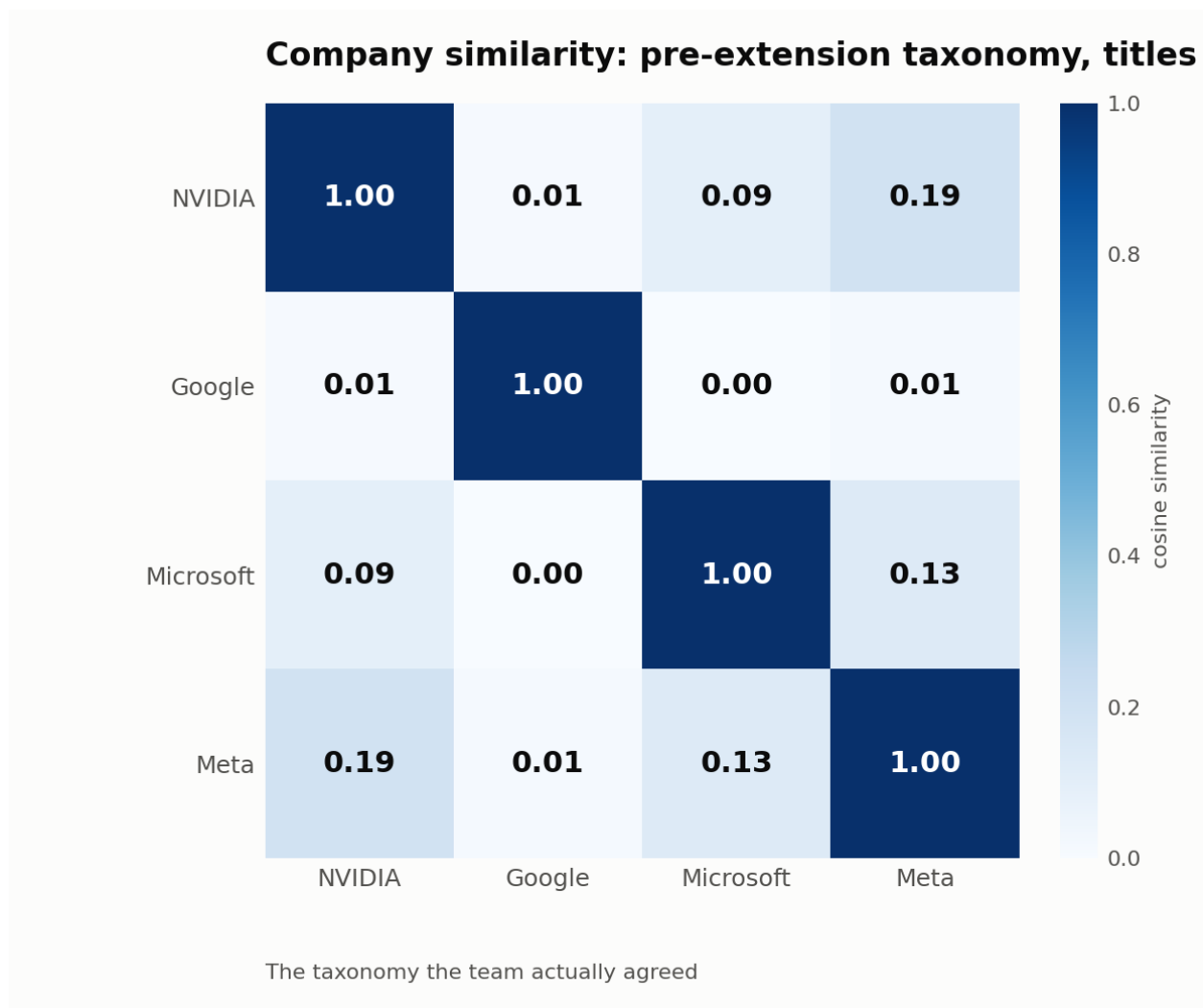


Figure 2. M2 - the pre-extension taxonomy the team actually agreed.

Pair	M1 extended	M2 pre-extension	Shift
NVIDIA - Microsoft	0.216	0.093	+0.123
NVIDIA - Google	0.075	0.009	+0.066
Google - Microsoft	0.051	0.003	+0.048
NVIDIA - Meta	0.147	0.195	-0.048
Microsoft - Meta	0.099	0.131	-0.033
Google - Meta	0.016	0.014	+0.002

The ranking flips. On M1 NVIDIA's closest peer is Microsoft at 0.216; on M2 it is Meta at 0.195, and Microsoft drops to 0.093. Changing the taxonomy moves that pair by 0.123, which is larger than most of the scores in the matrix. A measurement whose ordering depends on that choice is not yet a measurement.

3 The matrix that works

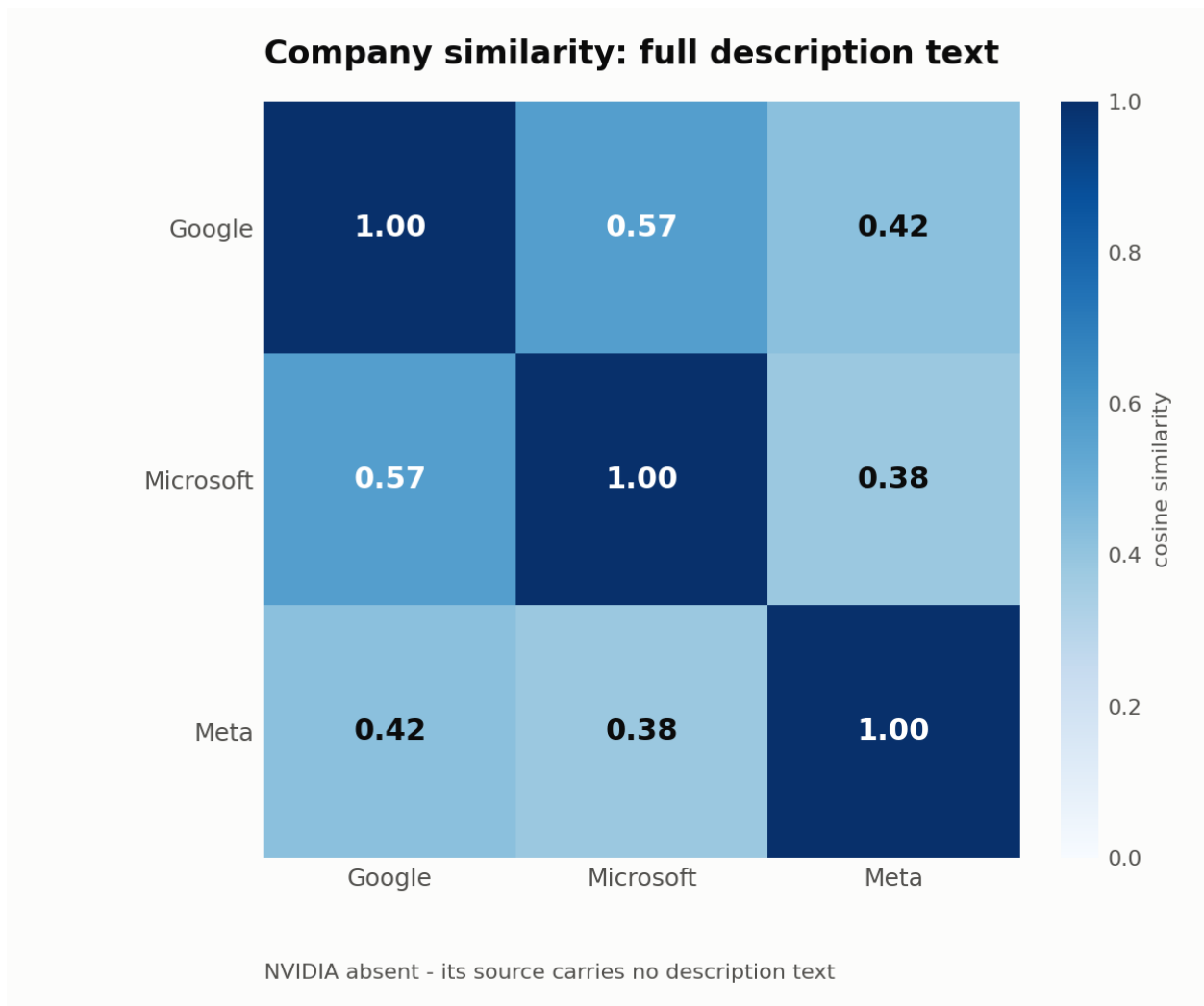


Figure 3. M3 - full description text. Scores an order of magnitude higher than the title matrices.

Google and Microsoft are the most alike at 0.57, which is what one would expect of two large cloud-and-platform employers hiring similar profiles. Meta sits apart from both at 0.42 and 0.38. These scores are stable and usable.

NVIDIA cannot appear in this matrix at all. That is the cost, carried all the way from the Task 1 finding that NVIDIA's Terms of Service rule out collecting from its own careers site.

4 What drives the pairs

Pair	Skills in common	Top shared	Top divider
NVIDIA - Microsoft	5	Cloud Infrastructure, Cybersecurity, Embedded Systems	Solutions Architecture, Networking, Datacenter Systems
NVIDIA - Meta	12	Datacenter Systems, Machine Learning, Robotics	Solutions Architecture, Networking, Cloud Infrastructure
NVIDIA - Google	24	Solutions Architecture, Cloud Infrastructure, Networking	GCP, Solutions Architecture, Networking

Pair	Skills in common	Top shared	Top divider
Google - Microsoft	5	Cloud Infrastructure, Cybersecurity, Machine Learning	GCP, Azure, Business Development

One pattern is worth pointing out. NVIDIA and Google share the **most** skills, 24 of them, yet score the **lowest** similarity at 0.075. Cosine similarity is driven by the weighted profile rather than the overlap count: the two companies touch many of the same skills at very different intensities. GCP dominates Google's vector; Solutions Architecture and Networking dominate NVIDIA's. Counting shared skills and measuring similarity are different questions, and it is worth saying which one a stakeholder is actually asking.

5 What can be concluded

Conclusion	Confidence
Google and Microsoft are the most similar pair with usable data, 0.57	High. Full text, stable across runs
NVIDIA is unlike the other three	Medium. True on every matrix, but NVIDIA's vector is built from titles while the others' best data is text
NVIDIA's closest peer	Not determined. M1 says Microsoft, M2 says Meta
Any precise similarity value involving NVIDIA	Low. Sparse vectors, plus a taxonomy bias worth up to 0.12

6 Limitations and what the team must do

#	Limitation
1	Title vectors are too sparse. One skill per posting produces near-orthogonal vectors. The low scores are a property of the data, not evidence that the companies are unrelated.
2	The taxonomy bias from Task 6 propagates here, worth up to 0.123 on a pair.
3	NVIDIA is absent from the only trustworthy matrix.
4	Sample sizes are unequal: 1,745 and 1,227 against 209 and 182.
5	No temporal dimension. The shared config lists trends as a similarity input; three of four datasets have no usable dates.

1. **Do not publish a four-company similarity matrix yet.** It would be read as a finding and it is not stable. This is the most important sentence in this document.
2. **Every member extends the taxonomy for their own company** - the same ask as Task 6, and the precondition for this task.
3. **Settle titles versus descriptions.** M3 works and M1 does not, and the only difference between them is the input text.

Files

File	Contents
data/similarity/similarity_M1_extended_titles.csv	4x4, extended taxonomy, titles
data/similarity/similarity_M2_original_titles.csv	4x4, pre-extension taxonomy, titles
data/similarity/similarity_M3_extended_fulltext.csv	3x3, full description text
data/similarity/pair_drivers.csv	Shared and dividing skills for every pair

File	Contents
data/similarity/matrix_comparison.csv	How far the taxonomy choice moves each pair
figures/	The three heatmaps above
similarity.py	The analysis, re-runnable from a clean checkout

```
cd work/task-8/nvidia-nayab-khalid
python similarity.py
```

Declaration	No personal or sensitive data is present in any output. No new data was collected; the inputs are the four members' committed datasets and both versions of the shared taxonomy.
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Date	20 September 2026